

Lab 1:

Statistics X Dependencies X Dependents X Processes X dvdrental/postgres@PostgreSQL 18*

dvdrental/postgres@PostgreSQL 18

Query Query History Scratch Pad X

```

1  -- LAB 1: Inspecting Table Size
2
3  -- Check table vs index size
4  SELECT
5      relname AS object_name,
6      pg_size_pretty(pg_table_size(rel.oid)) AS data_size,
7      pg_size_pretty(pg_indexes_size(rel.oid)) AS index_size,
8      pg_size_pretty(pg_total_relation_size(rel.oid)) AS total_size
9  FROM pg_class rel
10 JOIN pg_namespace n ON n.oid = rel.relnamespace
11 WHERE rel.relkind = 'r'
12       AND n.nspname = 'public'
13 ORDER BY pg_total_relation_size(rel.oid) DESC;

```

Data Output Messages Notifications

Showing rows: 1 to 17 Page No: 1 of 1

	object_name name	data_size text	index_size text	total_size text
1	rental	1232 kB	1120 kB	2352 kB
2	payment	896 kB	920 kB	1816 kB
3	film	736 kB	200 kB	936 kB
4	film_actor	272 kB	216 kB	488 kB
5	inventory	232 kB	208 kB	440 kB
6	customer	96 kB	112 kB	208 kB
7	address	88 kB	64 kB	152 kB
8	film_category	72 kB	40 kB	112 kB
9	city	64 kB	48 kB	112 kB
10	actor	40 kB	32 kB	72 kB
11	store	8192 bytes	32 kB	40 kB
12	staff	16 kB	16 kB	32 kB
13	language	8192 bytes	16 kB	24 kB
14	category	8192 bytes	16 kB	24 kB
15	country	8192 bytes	16 kB	24 kB
16	secure_orders	8192 bytes	16 kB	24 kB
17	rental_log	8192 bytes	8192 bytes	16 kB

Table: + rental (data_size = 1232 kB, index_size = 1120 kB)
+ payment (data_size = 896 kB, index_size = 920kB)

Lab 2: Step 1: Check current size

```
15  -- LAB 2: The Impact of a New Index
16
17  -- Step 1: Check current size of payment indexes
18  SELECT pg_size_pretty(pg_indexes_size('payment'));
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	pg_size_pretty text
1	920 kB

Step 2 + 3: Create a heavy index on payment_date then check again

```
20  -- Step 2: Create a heavy index on payment_date
21  CREATE INDEX idx_payment_date_long
22  ON payment (payment_date, amount, customer_id);
23  -- Step 3: Check size again
24  SELECT pg_size_pretty(pg_indexes_size('payment'));
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	pg_size_pretty text
1	1488 kB

- New index increased the index storage by approximately 568kB

Lab 3:

```
26 -- LAB 3: Partial Index Practice
27
28 DROP INDEX IF EXISTS idx_expensive_films;
29
30 CREATE INDEX idx_expensive_films
31 ON film (replacement_cost)
32 WHERE replacement_cost > 25.00;
33
34 SELECT relname, pg_size_pretty(pg_relation_size(oid))
35 FROM pg_class
36 WHERE relname = 'idx_expensive_films';
37
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	relname name	pg_size_pretty text
1	idx_expensive_fil...	16 kB

```
39 CREATE INDEX idx_full_replacement_cost
40 ON film (replacement_cost);
41
42
43 SELECT
44     relname AS index_name,
45     pg_size_pretty(pg_relation_size(oid)) AS index_size
46 FROM pg_class
47 WHERE relname IN (
48     'idx_expensive_films',
49     'idx_full_replacement_cost'
50 );
```

Data Output Messages Notifications

Showing rows: 1 to 2 of 1

	index_name name	index_size text
1	idx_expensive_films	16 kB
2	idx_full_replacement_c...	40 kB

- Partial index `idx_expensive_films` use 16 kB less than the full index `replacement_cost` 40 kB. -> because partial index only stores entries for films with a replacement cost higher than 25